

Generation Interconnection System Impact Study Report

For Queue Project Q18-0914

Gateway 138 kV
December 18, 2019

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q18-0914. The Interconnection Customer proposes a 203.5 MW wind facility interconnecting at the Gateway 138 kV substation in the New Jersey area. The request was received on January 08, 2018. The study evaluated system intact and single contingency conditions using the regional planning case, including 267 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 8 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$12,664,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Colorado / Yahara River 138kV	Russell Energy Center / Fair Lawn Substation 138kV	144.2	271	6.4
Coronado / Grandview 138kV line	Gateway / Blanchardville 138kV	132.0	424	100.0
Rosita Creek / Grandview 138kV	Blanchardville Substation / Gateway 138kV	130.0	444	100.0
Roman Nose / Glenore 138kV	Glenmore / Plaza Walmart 138kV line	118.2	267	45.1
Brownfield City / Plaza Walmart 138kV	Roman Nose / Glenmore 138kV line	117.4	297	53.3
Glenmore / Plaza Walmart 138kV	Roman Nose / Glenmore 138kV	116.3	308	53.3
Glenmore / Blanchardville Substation 138kV	Grandview / Rosita Creek 138kV	115.3	490	100.0
Gateway Substation / Blanchardville 138kV	Grandview / Rosita Crk 138kV	114.5	505	100.0

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Colorado Substation / Yahara River 138kV	Rebuild line with higher-capacity conductor	\$766,000	17% (\$127,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Coronado / Grandview 138kV line	Rebuild line with higher-capacity conductor	\$1,949,000	33% (\$651,000)
Rosita Creek / Grandview 138kV	Rebuild line with higher-capacity conductor	\$8,514,000	33% (\$2,841,000)
Roman Nose / Glenore 138kV	Rebuild line with higher-capacity conductor	\$7,231,000	39% (\$2,806,000)
Brownfield City / Plaza Walmart 138kV	Rebuild line with higher-capacity conductor	\$4,082,000	39% (\$1,580,000)
Glenmore / Plaza Walmart 138kV	Rebuild line with higher-capacity conductor	\$2,695,000	39% (\$1,043,000)
Glenmore / Blanchardville Substation 138kV	Rebuild line with higher-capacity conductor	\$3,879,000	33% (\$1,295,000)
Gateway Substation / Blanchardville 138kV	Rebuild line with higher-capacity conductor	\$6,955,000	33% (\$2,321,000)
	Total allocated to project		\$12,664,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.